DEPARMENT OF INFORMATION TECHNOLOGY
FACULTY OF ENGINEERING AND TECHNOLOGY

IT25-425 P-II
COMPUTER NETWORKS
(MORNING & EVENING)

LAB#2

UNDERSTANDING & SETTING UP CROSSOVER CABLE

Name: _____ Roll No: _____

Score: _____ Signature of the Lab Tutor _____ Date: _____

Objectives:

- Build a straight-through Ethernet patch cable to T568-B (OR T568-A) standards for connection from workstation to hub/switch or patch panel to hub/switch.

Background:

In this lab you will learn how to build a Category 5e or 6 (CAT 5e or 6) Unshielded Twisted Pair (UTP) Ethernet network patch cable (or patch cord) and test it for good connections (continuity) and correct pinouts (correct color of wire on the right pin). This will be a 4-pair (8-wires) "straight through" cable which means that the color of wire on pin 1 on one end of the cable will be the same as pin 1 on the other end. Pin 2 will be the same as pin 2 and so on. It will be wired to TIA/EIA-568-B or A standard for 10BASE-T Ethernet which determines what color wire is on each pin. T568-B (also called AT&T specification) is more common, but many installations are also wired to T568-A

This patch cable will conform to the structured cabling standards and is considered to be part of the "horizontal" cabling which is limited to 99 meters total between workstation and hub or switch. It can be used in a workstation area to connect the workstation NIC to the wall plate data jack or it can be used in the wiring closet to connect the patch panel (horizontal cross connect) to an Ethernet hub or switch. Patch cables are wired straight thru since the cable from the workstation to the hub or switch is normally crossed over automatically at the switch or the hub. Note that the ports on most hubs have an X next to them. This means the send and receive pairs will be crossed when the cabling reaches the switch. The pinouts will be T568-B and all 8 conductors (wires) should be terminated with RJ-45 modular connectors (only 4 of the 8 wires are used for 10/100/1000 BASE-T Ethernet, all 8 are used for 1000BASE-T Ethernet).

Tools / Preparation:

Prior to starting the lab, the teacher or lab assistant should have a spool of Cat 5e or 6 Unshielded Twisted Pair (UTP) cable, RJ-45 (8-pin) connectors, an RJ-45 crimping tool and an Ethernet / RJ-45 continuity tester available. Work individually or in teams. The following resources will be required:

- Two to three foot length of Cat 5e or 6 cabling (one per person or one per team)
- Four RJ-45 connectors (two extra for spares)
- RJ-45 crimping tools to attach the RJ-45 connectors to the cable ends
- Ethernet cabling continuity tester which can test straight-thru or crossover type cables (T568-A or T568-B).
- Wire cutters

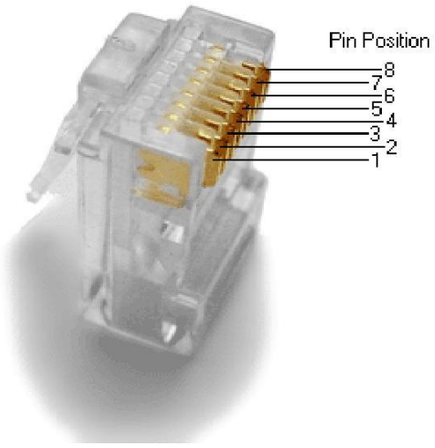
Step 1 - Cabling Information.

Explanation: Instructions are provided here for building a T568-A or T568-B cable. Either can be used as long as all connections (pinouts) from the workstation to the wiring closet and terminating electronics (hubs or switches) are consistent. If cables are to be built for an existing network it is important to keep the same standard as already exists (either T568-A or B). A patch cable that is wired "straight through" will have the same color of wire on the same pin (1 -- 8) at both ends. A straight through patch cable (T568-A or B) can be used to connect a PC workstation to a wall plate in a work area or it can be used to connect from a patch panel in a wiring closet to a hub or a switch. A PC can also be connected directly to a port on a hub or switch with this cable. If a cable will be used to connect from an "uplink" port on one hub to a "crossover" front port on another hub then a straight through cable should be used

Step 2 - Create a T568-A crossover patch panel cable.

Task: Use the following tables and diagrams and steps to create a T568-B patch panel cable.

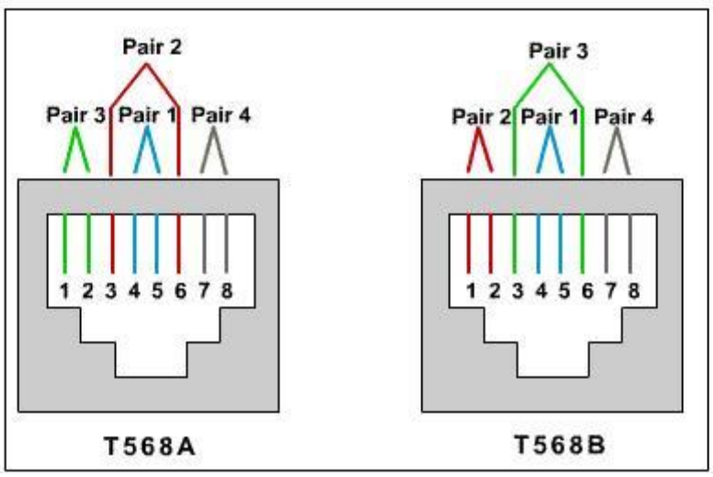
Explanation: Both cable ends should be wired the same when looking at the conductors. Only four wires are used with 10BASE-T or 100BASE-TX ot 1000BASE-T Ethernet:

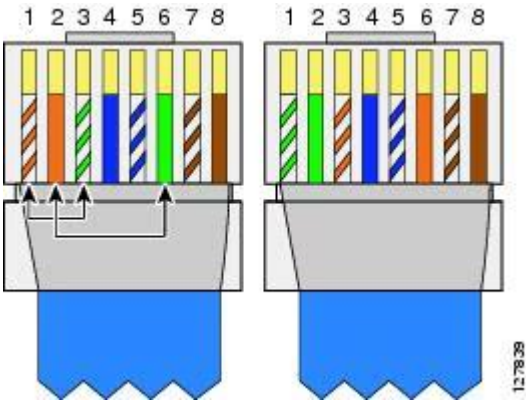


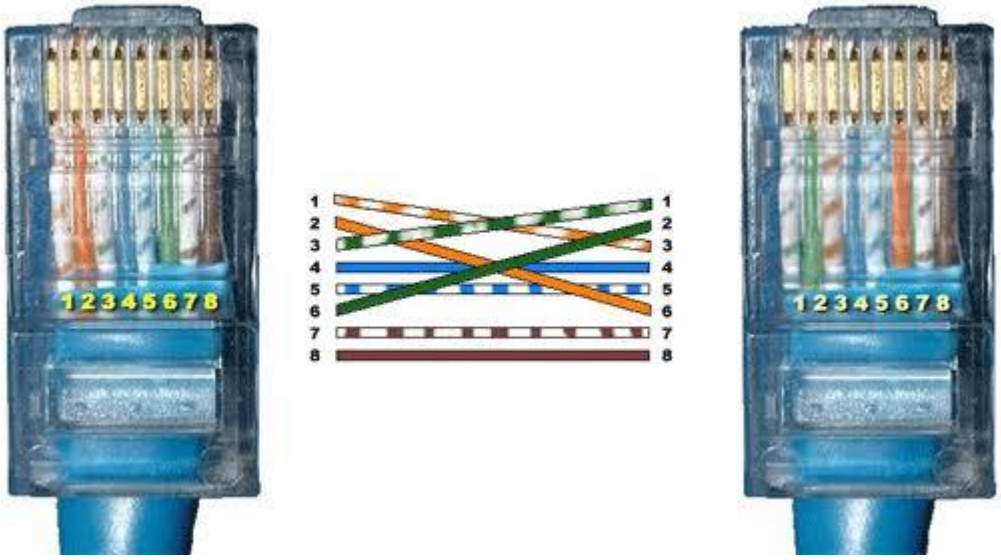


Pin#	Pair#	Function	Wire Color	Used with 10/100 BASE-T Ethernet?	Used with 100 BASE-T4 and 1000 BASE-T Ethernet?
1	2	Transmit	White/Orange	Yes	Yes
2	2	Transmit	Orange/White	Yes	Yes
3	3	Receive	White/Green	Yes	Yes
4	1	Not used	Blue/White	No	Yes
5	1	Not used	White/Blue	No	Yes
6	3	Receive	Green/White	Yes	Yes
7	4	Not used	White/Brown	No	Yes
8	4	Not used	Brown/White	No	Yes

Diagram showing both T568-A and T568-B cabling wire colors







1. Determine the distance between devices, or device and plug, then add at least 12" to it. The maximum length for this cord is 3 m; standard lengths are 6' and 10'.
2. Cut a piece of stranded Cat 5e or 6 unshielded twisted-pair cable to the determined length. You will use stranded cable for patch cables because it is more durable when bent repeatedly. Solid wire is fine for cable runs that are punched down into jacks.
3. Strip 2" of jacket off of one end of the cable.
4. Hold the 4 pairs of twisted cables tightly where jacket was cut away, then reorganize the cable pairs into the order of the 568-B wiring standard. Take care to maintain the twists since this provides noise cancellation. (orange pair, green pair, blue pair, brown pair).
5. Hold the jacket and cable in one hand, untwist a short length of the green and blue pairs, and reorder

them to reflect the 568-A wiring color scheme. Untwist and order the rest of the wire pairs according to the color scheme.

6. Flatten, straighten, and line up the wires, then trim them in a straight line to within 1/2" - 3/4" from the edge of the jacket. Be sure not to let go of the jacket and the wires, which are now in order! You should minimize the length of untwisted wires because overly-long sections that are near connectors are a primary source of electrical noise.
7. Place an RJ-45 plug on the end of the cable, with the prong on the underside and the orange pair to the left side of the connector.
8. Gently push the plug onto wires until you can see the copper ends of the wires through the end of the plug. Make sure the end of the jacket is inside the plug and all wires are in the correct order. If the jacket is not inside the plug, it will not be properly strain relieved and will eventually cause problems. If everything is correct, crimp the plug hard enough to force the contacts through the insulation on the wires, thus completing the conducting path.
9. Repeat steps 3-8 to terminate the other end of the cable, using the same scheme to finish the straight through cable.
10. Test the finished cable and have the instructor check it. How can you tell if your cable is functioning properly?

Review Questions

Question 1 What is a crossover cable?
Answer:

_____

_____

_____

Question 2: Which wiring standards does it use?
Answer:

_____

_____

_____

Question3: When do you need a Crossover Cable?
Answer:

_____

_____

_____

Note: Submit the handout before next lab